

$$7.5 + 0.5 = 8$$

PART B
(Total Marks: 2*5=10)

Instructions for PART B: This is a short-answer type question. You have to answer in the space given.

1. List and briefly explain two key assumptions of linear regression.

1. Linearity: all features are linearly correlated
2. Independence: all features are independent of each other

2. What does overfitting mean in the context of regression models? Mention one method to prevent it.

Overfitting: train MSE ↓ but test MSE ↑, not able to predict for unseen data properly (performs poorly on test data). To prevent it, validation split can be used to reduce overfitting.

3. Why is feature scaling important in logistic regression but not always in linear regression?

Scaling is imp. in logistic reg. because without scaling, sigmoid/softmax will give unprecedented outputs since it squashes the overall values in between 0 and 1.

4. What role do residuals play in checking the validity of a linear regression model?

R^2 score = $1 - \frac{SS_{res}}{SS_{total}}$, SS_{res} = residual sum of squares = $\sum (y_i - \hat{y})^2$, SS_{total} = mean sum of squares = $\sum (y_i - \bar{y})^2$. If $R^2 = 1 \Rightarrow SS_{res} = 0 \Rightarrow$ all variance is captured. If $R^2 = 0 \Rightarrow SS_{res} = SS_{total} \Rightarrow$ model not appropriate.

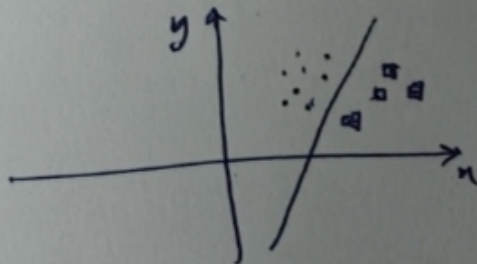
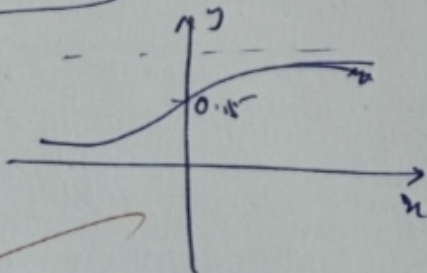
5. In logistic regression, how is the decision boundary defined mathematically?

$$w_0 x_0 + w_1 x_1 + w_2 x_2 + \dots + w_n x_n - y = 0 \rightarrow \text{decision boundary}$$

$$w^T x \geq 0 \Rightarrow g(w^T x) \geq 0.5 \Rightarrow \text{class 1}$$

$$w^T x < 0 \Rightarrow g(w^T x) < 0.5 \Rightarrow \text{class 0}$$

sigmoid



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